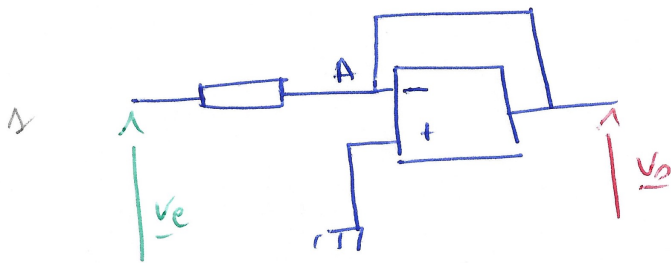


Filtre actif

① A HF :



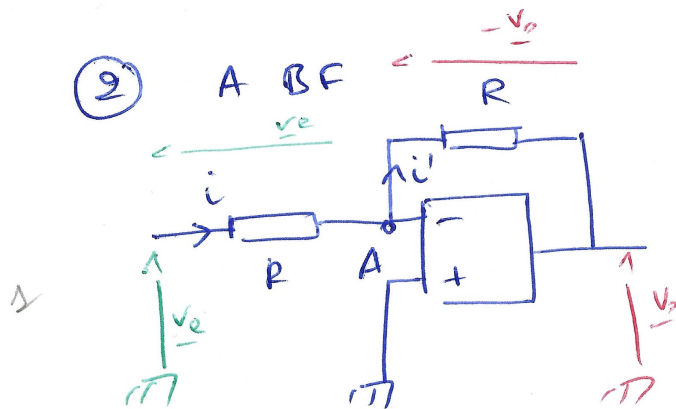
R est court-circuitée.

$\varepsilon = 0$ (AOP idéal à régime lin.)

d'où $\underline{V_A} = 0 = \underline{V_o}$

$$\Rightarrow \underline{H} = 0$$

② A BF



$\underline{V_A} = 0$ (même raison).

$$i = i'$$

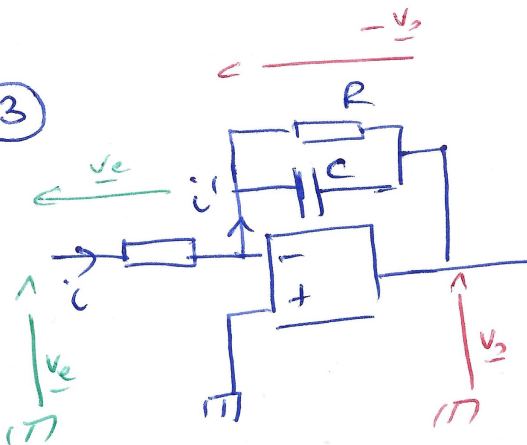
$$\frac{V_e}{R} = -\frac{V_o}{R}$$

$$\underline{V_o} = -\underline{V_e}$$

$$\underline{H} = -1$$

passer-bas.

③



$$i = i'$$

$$\frac{V_e}{R} = -\frac{V_o}{Z_{||}}$$

$$\frac{V_e}{R} = -\frac{V_o}{R} \left(\frac{1}{R} + j\omega C \right)$$

$$\underline{V_o} = -\frac{V_e}{R \left(\frac{1}{R} + j\omega C \right)}$$

$$\underline{H} = \frac{V_o}{V_e} = -\frac{1}{1 + jRC\omega}$$

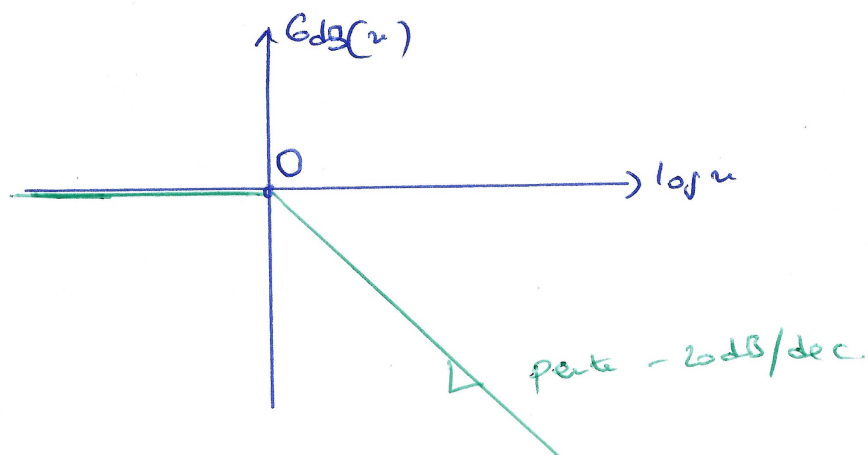
④ Or pole $\omega_0 = \frac{1}{RC}$

1 $\underline{H} = \frac{-1}{1+j\omega} \quad \omega = \frac{\omega}{\omega_0}$

⑤ $G = |\underline{H}| = \frac{1}{\sqrt{1+\omega^2}}$

$\omega \rightarrow 0$: $\underline{H} \rightarrow -1$ $G(\omega) \rightarrow 1$ $G_{dB} = 20 \log G(\omega) \rightarrow 0$

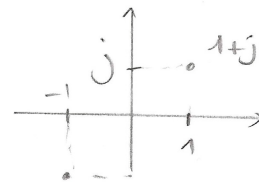
2 $\omega \rightarrow +\infty$ $\underline{H} \sim -\frac{1}{j\omega} = \frac{j}{\omega}$ $G(\omega) \rightarrow 0$ $G_{dB} = 20 \log\left(\frac{1}{\omega}\right) = -20 \log \omega$



⑥

$\omega \rightarrow 0$ $\underline{H} \rightarrow -1$ $\varphi(\omega) \rightarrow \pi$

$\omega \rightarrow +\infty$ $\underline{H} \sim \frac{j}{\omega}$ $\varphi(\omega) \rightarrow +\pi/2$

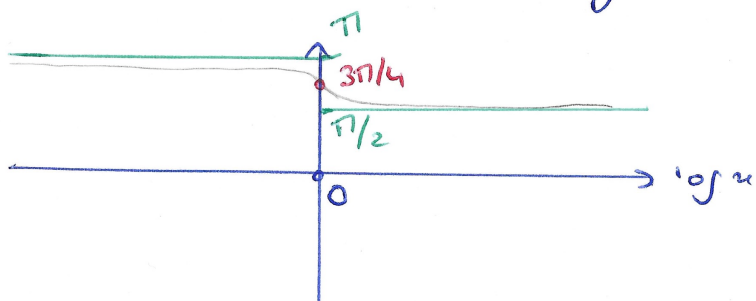


2 $\omega = 1$ $\underline{H}(1) = \frac{-1}{1+j}$

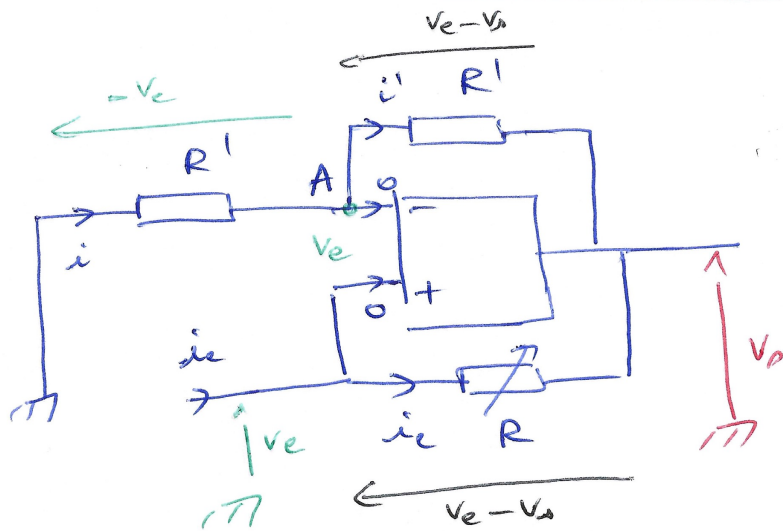
$\arg\left(\frac{-1}{1+j}\right) = -\arg(1+j) + \arg(-1)$

$= \pi - \pi/4$

$= 3\pi/4$



Droble d'impédance



- ① loi des nœuds en A : $i = i'$ (AO idéal $i_+ = i_- = 0$)

AO idéal en régime linéaire $\varepsilon = 0 \Rightarrow V_A = V_e$.

$$i = i'$$

$$\frac{-V_e}{R'} = \frac{V_e - V_d}{R'}$$

$$-V_e = V_e - V_d$$

$$\boxed{V_d = 2V_e}$$

- ② L'impédance d'entrée est définie par

$$\underline{Z_e} = \frac{\underline{V_e}}{\underline{i_e}}$$

or $\underline{i_e} = \frac{\underline{V_e} - \underline{V_d}}{R}$ en RPF

$$= \frac{\underline{V_e} - 2\underline{V_e}}{R}$$

donc $\underline{Z_e} = \frac{\underline{V_e}}{\underline{i_e}} = \boxed{-R}$

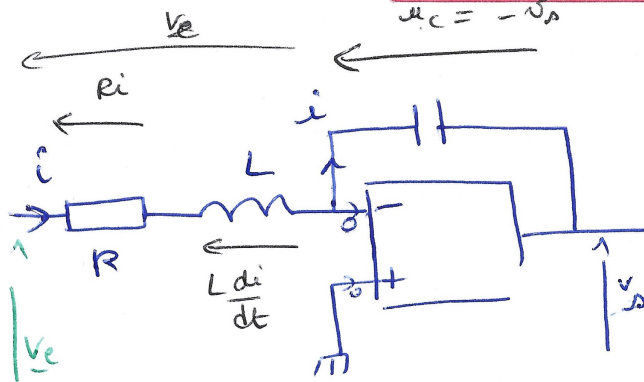
$$= -\frac{\underline{V_e}}{R}$$

- ③ Ce montage est surprenant car l'impédance d'entrée correspond à 1 "résistance négative".

Réponse à 1 échelon

$$u_c = -v_s$$

16



$$i = C \frac{dv_s}{dt}$$

$$= -C \frac{dv_s}{dt}$$

$$v_e = R i + L \frac{di}{dt}$$

$$v_e = -RC \frac{dv_s}{dt} - LC \frac{d^2 v_s}{dt^2}$$

2

①

$$\frac{di}{dt} + \frac{R}{L} i = \frac{v_e}{L} = \frac{E}{L}$$

②

$$i(t) = A e^{-t/\tau} + i_p$$

$$= A e^{-t/\tau} + \frac{E}{R}$$

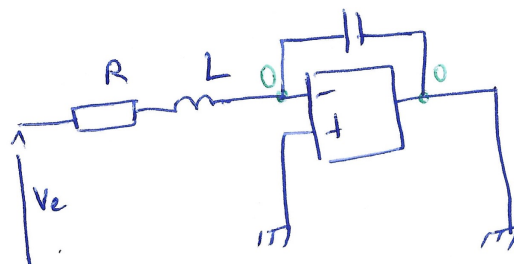
$$\tau = \frac{L}{R}$$

$$i_p = \frac{E}{R}$$

CI:

$$\text{à } t=0 \quad v_s=0$$

$$t=0^- \quad v_e=0$$



2

$$i(0)=0$$

$$i(t) = \frac{E}{R} (1 - e^{-t/\tau})$$

$$v_s = -u_c$$

$$i = C \frac{dv_s}{dt} = -C \frac{dv_s}{dt}$$

$$-\frac{i(t)}{C} = \frac{dv_s}{dt}$$

$$V_o(t) = - \int_0^t \frac{j(t')}{C} dt'$$

$$= - \int_0^t \frac{E}{RC} (1 - e^{-t'/\tau}) dt'$$

$$= - \frac{E}{RC} t + \left(-\tau \right) \frac{E}{RC} (e^{-t/\tau} - 1)$$

$$= - \frac{E}{RC} t - \frac{L}{R} \cdot \frac{E}{RC} (e^{-t/\tau} - 1)$$

2

$$V_o(t) = - \frac{E}{RC} \left[t + \frac{L}{R} (e^{-t/\tau} - 1) \right]$$